

Head-Up Displays in Urban Partially Automated Driving: Effects of Using Augmented Reality*

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Abstract— In partially automated driving the vehicle takes over lateral and longitudinal control in specific situations. The driver acts as a supervisor of the driving automation system, and must be able to intervene immediately if functional limits are reached. Therefore, the driver needs to be aware of actual and planned system actions. Due to the complexity of the urban traffic environment when compared with that of well-structured highways, optimized driver-vehicle interaction is essential. The choice of a suitable display seems to be a key factor, since most information is perceived visually while driving. A promising method for supporting the driver is an Augmented Reality Head-Up Display (ARHUD). An ARHUD visualizes information at its actual location in the primary field of view in contrast to a conventional Head-Up Display (HUD). To investigate the potential of an ARHUD in comparison with a HUD, 30 participants took part in a driving simulator study experiencing a partially automated journey supported by an ARHUD or a HUD in a within-subjects design. Results show significant differences in gaze behavior for the two display concepts. The ARHUD results significantly in higher trust and lower subjective workload compared to the HUD. However, no significant differences exist in usability.

I. INTRODUCTION

A. Partially Automated Driving

In recent years, partially automated driving became the state of the art in production vehicles. In partially automated driving the vehicle takes over lateral and longitudinal control. According to [1], the driver has to act continuously as a supervisor of the driving automation system, and must be able to resume control of the vehicle immediately if functional limits are reached, in order to keep the vehicle operating safely. Functional limits do not necessarily have to be prompted by the vehicle. The current operational design domain (ODD) of partially automated driving is solely on highways because an urban traffic environment is much more complex by comparison with well-structured highways. In urban areas, drivers encounter high complexity caused by various reasons as the diversity of road users and high information density [2]. In addition, passenger seem to trust automated vehicles less in urban traffic than on highways or rural roads [3].

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B. Head-Up Displays

To support the driver in his monitoring task by presenting information on the system status, information about the actual and planned maneuver is necessary [4, 5, 6]. The high information density in urban areas and the high level of visual perception required while driving [7] lead to a further investigation on how the driver could be supported visually in his monitoring task.

Previous studies have already shown positive effects of using Head-Up Displays (HUD) in the Human-Machine-Interface (HMI) design for safe, efficient and low-stress urban driving [2]. In automated driving HUDs have generally been used to communicate the system status, speed and automated maneuver [8, 9]. In general, HUDs are already viewed as a state-of-the-art application.

Augmented Reality Head-Up Displays (ARHUD) are a further development of the conventional HUD, which is characterized by expanding the real world with virtual information [10]. In contrast to a conventional HUD, an ARHUD visualizes information at the location in the primary field of view, where the information is needed in a specific situation. Potentially, this would seem to offer various advantages when using an ARHUD. Overall, it should reduce the driver's cognitive effort and increase driving safety and comfort [11]. Reference [12] showed that an augmentation of sensor data in the driver's field of view can enhance trust and acceptance in the automated vehicle. Further effects of reduced workload, reduced subjective distraction and a higher sense of safety in monitoring an adaptive cruise control (ACC) were observed for an ARHUD in comparison with an Instrument Cluster (IC) [13]. A further comparison of ARHUD and an IC in conditionally automated driving revealed that 80 % of the sample preferred the ARHUD concept and showed a higher understandability and usefulness of the ARHUD [14]. A study by [15] has shown that an augmented visualization of information in partially automated driving can help to increase system understanding and improve reaction speeds in the event of a system failure.

C. Objectives

Previous studies have shown the positive effects of using ARHUD and HUD for the driver. However, many of these studies did not consider the advantages of ARHUD relative to HUD, focusing rather on a comparison of ARHUD or HUD with the IC. In particular, in respect of urban partially automated driving, it is still a matter of debate as to whether

an ARHUD or a HUD is more supportive in monitoring the automated vehicle's maneuver. Therefore, this work investigates the effects of an ARHUD compared with a HUD on the gaze behavior and subjective perception while monitoring a partially automated driving system. We defined the following hypotheses for this study:

H1: Using ARHUD and HUD results in differences in gaze behavior.

H2: The ARHUD concept results in reduced subjective workload in contrast to the HUD concept.

H3: The ARHUD concept results in increased trust when compared to the HUD concept.

H4: The ARHUD concept results in improved usability compared to the HUD concept.

II. HMI DESIGN

The HUD concept consisted of a conventional HUD and an additional IC, while the ARHUD concept consisted of the ARHUD, elements of the HUD for a preview of the navigation and speed display and the IC. The display in the IC was equivalent in both concepts. The IC's visualization of system status and maneuvers was very similar to the visualization in the HUD presented below (see Fig. 1).

The HUD consisted of three sections (see Fig. 1). The left section showed the actual speed and speed limit, the middle section presented the status and maneuver of the automated driving system and the right section illustrated the navigation information. The speed and navigation information were taken from [2]. The system status and maneuver were based on the information concept of the UR:BAN project developed by [2] and adapted for partially automated driving by extending the icon for the lateral and longitudinal guidance according to [16].

The left and right sections of the HUD have also been visualized in the ARHUD concept to show the same amount of information in both concepts. We considered the following design guidelines in the development process of the ARHUD concept: superimposition of objects in the real environment and virtual elements in the ARHUD should be avoided [17]. Large, dominant visualizations and the visualization of a large amount of elements should also be avoided in order not to distract the driver but to direct his attention depending on the traffic situation. We used a boomerang chain to visualize the planned trajectory of the automated vehicle (see Fig. 2). A boomerang chain proves to be advantageous because it is perceived as a trajectory and

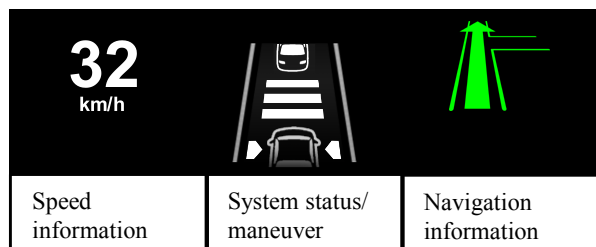


Fig. 1. Visualization in the HUD (Scenario: Following a front vehicle)

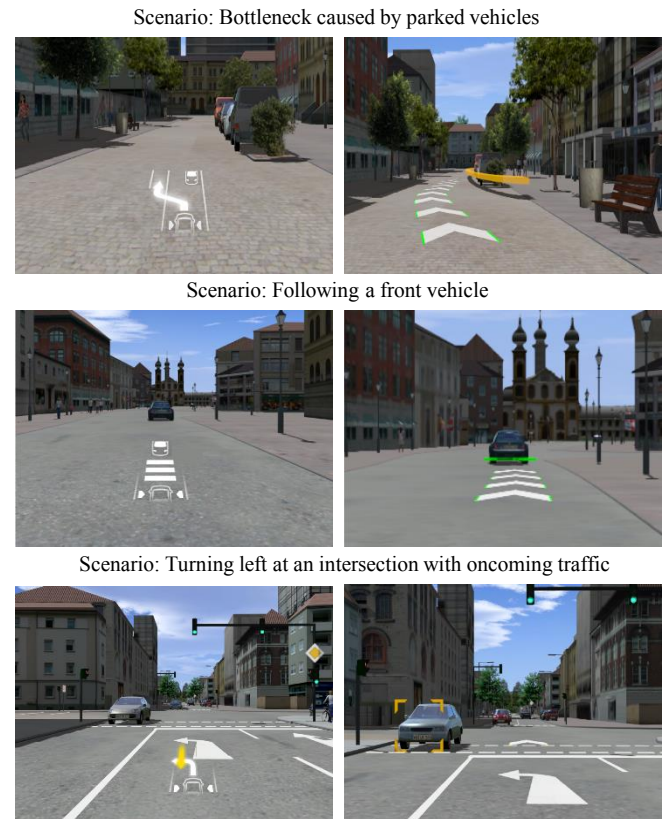


Fig. 2. Differences in the visualization of the system status and maneuver between the HUD (left) and the ARHUD (right)

has a lower risk of concealing real objects than, for example, a drawn-through arrow [18]. The presentation of the boomerang chain should be situational and not permanent [18], which is why we reduced the presentation to a single boomerang if the vehicle had to stop or if there was no explicit maneuver such as turning, changing lanes or following a front vehicle. The boomerang chain represented the control range for the case of following a front vehicle, as known from ACC icons. The arrow edges of the boomerang chain turned from green to yellow, when the automated vehicle had to decelerate. Oncoming or cross traffic is marked by the recommended visualization of a rectangular bracket around the objects [19]. A round bracket marks static and dynamic obstacles in the vehicle's lane. In addition to the marking, the visualization of round brackets should also clarify the lateral distance to the object when passing. Fig. 2 shows the different visualization of the system status and maneuver of the HUD and ARHUD concept for three specific scenarios.

III. METHOD

A. Participants

Thirty-seven participants took part in the experiment. Two data sets had to be excluded for technical reasons. Additionally five data sets were unusable because five participants had to stop the experiment because of simulation sickness. The remaining sample of 30 participants consisted of 33 % female and 67 % male participants. The average age

was 28.56 years, ranging from 21 to 60 years. Twenty-one participants had a technical work or study program background, whom we would expect to be open-minded in respect of technical applications. Thirteen participants had already used HUD. In contrast, only three participants had already experienced augmented reality applications in a nonautomotive context. A four years driver's license was the prerequisite for participation to avoid distortion of the results by novice drivers.

B. Experimental Design

We used a within-subjects design for the factor HMI (HUD and ARHUD) to minimize interindividual influences in the HMI evaluation. Each participant experienced each HMI concept in a separate experiment drive in a permuted order. Each experiment drive took about 14 minutes to avoid a decrease in vigilance due to the duration of monitoring of the partially automated driving system [6]. The instruction for the participants was to activate the partially automated driving system whenever it was available. After a one-minute manual drive, partial automation was available for the remaining 13 minutes with no take-over of control by the driver being necessary. The participants did not have to drive hands-on, as longer hands-off periods do not necessarily have a negative impact on the take-over performance [20]. During the automated drive, typical urban traffic scenarios such as bottlenecks, overtaking of cyclists, following vehicles in front, intersections with traffic lights e.g. were passed with a maximum speed of 50 km/h or 30 km/h (approx. 31.1/18.6 mph).

C. Dependent Variables

This work measured the gaze behavior as an objective variable and several subjective variables using questionnaires. The gaze behavior was operationalized by the gaze duration on the relevant areas of interest (AOIs) for monitoring the actions and maneuver of the automated vehicle. The gaze duration has been calculated as a percentage of the automated travel time. AOIs were road ahead, HUD, IC, rear mirror, left mirror and right mirror (see Fig. 3). The System Usability Scale questionnaire measured the usability of the HMI concepts [21]. We measured the subjective workload in monitoring the partially automated driving system using the NASA-TLX questionnaire [22] and the trust in the partially automated vehicle using the Trust in Automation questionnaire [23]. In order to obtain data about the information perception of the individual HMI concepts additional nine questions were asked. The questions were taken from [6] with regard to the information contents and time of information. The participants answered these questions on a 15-point Likert scale. All questionnaires were provided in German.

D. Apparatus

The study was conducted in the static driving simulator of the Chair of Ergonomics consisting of a BMW 6 series (E64) mock-up. Six projectors provided a 180° front view, and the view of the side and rear mirrors. An additional projector produced the HUD visualization. An installed

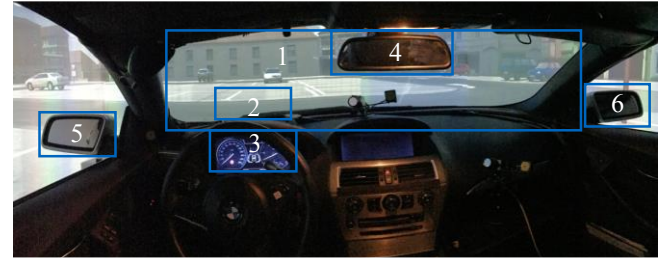


Fig. 3. AOIs marked blue in the vehicle mock-up; 1. Road ahead (without 2 and 4), 2. HUD, 3. IC, 4. Rear mirror, 5. Left mirror, 6. Right mirror

sound system created engine sounds and environmental noise. The software SILAB from the WIVW GmbH realized the driving simulation at 60 Hz. Three remote eye-tracking cameras from Smart Eye AB were installed in the vehicle interior to track drivers' eyes and gaze direction. Participants could activate the partial driving automation system via a button on the steering wheel. Deactivation of the driving automation system was possible by pressing the same button or by a steering or braking input.

E. Procedure

The experiment took about 90 minutes per participant. At the start, participants filled out a demographic questionnaire and signed the declaration of consent. Additionally, an introduction in written form was given explaining the driver's task during partially automated driving, the capabilities of the driving automation system and basic information about the various displays used for the different HMI concepts. The participants entered the mock-up, adjusted the seat, mirrors and steering wheel and conducted a familiarization drive to get to know the driving simulator. The familiarization drive consisted of five minutes manual and five minutes partially automated driving. Afterwards, we calibrated the eye-tracking system. After this, the two experiment drives took place in permuted order. Following each experiment drive, the participants had to complete a survey containing the questionnaires regarding trust, usability, subjective workload and information perception. The experiment was concluded with a follow-up questionnaire.

IV. RESULTS

A. Statistical Analysis

We conducted a two-tailed dependent t-test to analyze the gaze duration on different AOIs. To analyze differences in usability, subjective workload and trust a one-tailed dependent t-test was performed. We checked the normality assumption by applying the Shapiro-Wilk test. In the event of a violation of the normality distribution (usability; gaze duration on the road ahead, left/right/rear mirror), we also conducted a dependent t-test due to the sample size of 30 participants. For large sample sizes of at least 30 participants, the t-test is robust in the event of violations of the normality distribution [24]. A Wilcoxon signed-rank test was conducted to analyze the ordinal data of the subjective statements on the information perception. In all statistical tests, a significance level of 0.05 was applied. Cohen's

benchmark was used to classify the effect sizes [25]. We used MATLAB and MS Excel to prepare the data and JASP to carry out the statistical analysis.

B. Eye-Tracking

The analysis of the eye-tracking data showed an overall percentage of valid gaze data within an AOI of about 90 % for the automated drive with the HUD and ARHUD (see Table I). Reference [26] describes data availability of over 85 %, as in this case, as good. No participant had less than 70 % of valid data within the AOIs, which would have lead to the exclusion of the participant's eye tracking data from this sample.

The overall gaze duration on the AOIs calculated as a percentage of the partially automated drive is presented in Table II. A t-test for the AOI road ahead indicated a medium significant effect. When using the ARHUD, drivers looked significantly more at the road ahead than when using the HUD ($\Delta M = 4.25 \%$, $t(29) = 4.336$, $p < 0.001$, $d = 0.792$, two-tailed). The ARHUD concept in comparison to the HUD concept revealed a significantly lower percentage of the gaze duration on the AOI HUD ($\Delta M = 5.88 \%$, $t(29) = -5.931$, $p < 0.001$, $d = 1.083$, two-tailed), which is identified as a large significant effect. The drivers spent a significantly longer time looking at the IC, when using the ARHUD instead of the HUD ($\Delta M = 0.94 \%$, $t(29) = 2.718$, $p = 0.011$, $d = 0.496$, two-tailed) with a medium effect size. There was no significant effect for the AOIs rear mirror, left mirror and right mirror. In total, mirrors only accounted for a percentage of less than 2 % of the overall gaze duration. In summary, hypothesis 1 can be accepted.

C. Subjective Ratings

Fig. 4 shows the boxplots of the overall workload index (OWI) of the NASA-TLX score, the trust score and the SUS score for both HMI concepts. We found a small effect for the OWI, which describes the subjective workload ($t(29) = 1.712$, $p = 0.049$, $d = 0.313$, one-tailed). The OWI of the ARHUD ($M = 34$, $SD = 18.19$) was significantly lower than that of the HUD ($M = 36.79$, $SD = 16.47$), which leads to the acceptance of hypothesis 2.

Using the ARHUD ($M = 3.81$, $SD = 0.51$) led to a

TABLE I. AVAILABILITY OF GAZE DATA DURING THE PARTIALLY AUTOMATED DRIVE [N = 30]

Percentage of gaze data	HMI concept	
	HUD	ARHUD
Within AOI [%]	90.50	89.70
Outside AOI [%]	1.18	1.19
Missing [%]	8.32	9.11

TABLE II. OVERALL GAZE DURATION ON THE DIFFERENT AOIS DURING THE PARTIALLY AUTOMATED DRIVE [N = 30]

AOI	HUD		ARHUD	
	M [%]	SD [%]	M [%]	SD [%]
Road ahead	68.00	10.50	72.25	9.38
HUD	18.14	10.51	12.26	7.17
IC	2.52	1.32	3.46	1.88
Rear mirror	0.75	0.88	0.68	0.52
Left mirror	0.81	0.52	0.77	0.58
Right mirror	0.28	0.38	0.29	0.37

significantly higher trust in the automated vehicle compared to that when using the HUD ($M = 3.69$, $SD = 0.46$) ($t(29) = -2.122$, $p = 0.021$, $d = 0.387$, one-tailed), which is identified as a small effect. Therefore, hypothesis 3 can be accepted.

The statistical analysis showed no significant difference for the SUS scores ($t(29) = -1.367$, $p = 0.091$, one-tailed). Interpreted by [27] both HMI concepts revealed good usability (HUD: $M = 80.17$, $SD = 10.32$; ARHUD: $M = 83.58$, $SD = 14.01$), with a trend towards excellent for the ARHUD (see Fig. 5). Due to this result, hypothesis 4 is rejected.

The results of the questionnaire about the information perception of the HMI concepts rated on a 15-point Likert scale are presented in Table III. The rating indicated no significant differences, except for the fifth statement. The rating of the participants revealed a large effect, which was significantly lower for the HUD ($Mdn = 2$) than for the ARHUD ($Mdn = 3$) in respect of the statement "I have to

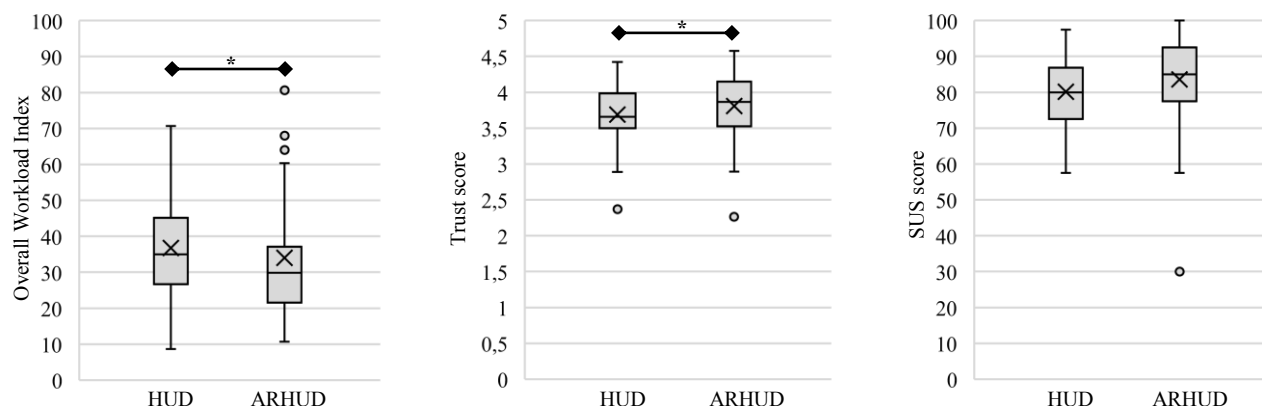


Fig. 4. Boxplots [n = 30] of the OWI of the NASA-TLX score (left), Trust Score (middle) and SUS score (right) for the HUD and ARHUD concept (* $p < 0.05$); x shows the mean value

remember a lot of details to understand the visualization in the AR(HUD)", $z = -3.098$, $p = 0.002$, $r = 0.566$, two-tailed.

V. DISCUSSION

This study showed a significant difference in the gaze behavior when using an ARHUD and HUD. The significant differences in the overall gaze duration on the AOIs HUD and road ahead could be explained by the fact that significantly more information is displayed in the HUD for the HUD concept case. By contrast, the ARHUD displays more information directly within the environment that corresponds to the AOI road ahead. The results therefore indicate a shift of attention to the information displayed. This assumption would fit in with a previous study, according to which an augmented reality presentation can direct attention to marked objects or make them more recognizable [28]. This allows the driver to react more quickly [28] and is therefore particularly advantageous in critical situations. The significant difference in respect of the AOI IC means a lower eyes-off-road time when using the HUD concept. This could relate to the fact that in the HUD concept the presentation of information about the system status and maneuver in the IC is very similar to the presentation in the HUD. As a result, participants may not find it a benefit to look additionally at the IC. In contrast, the presentation of information in the ARHUD differs considerably from that in the IC. The gaze at the IC would then serve as a kind of safety gaze, to ensure that no relevant system information was overlooked in the ARHUD. This assumption would fit in with [29] according to which gaze behavior must first adapt to new systems. Since the ARHUD represents the novel system in this work, a lack of experience with the ARHUD could be the reason for the greater focus on the IC.

The statistical analysis revealed a significantly lower subjective workload for the ARHUD concept despite its being used for the first time. This leads to the assumption that the information in the ARHUD resulted in less interpretation effort for the participants, which could lead to a higher attention retention. This result is particularly remarkable because no participant had experience with augmented reality in the vehicle context, while 13 subjects were accustomed to using HUD.

The results of trust in the automated vehicle support the results of [12], according to which an augmented visualization of sensor data in the driver's field of view can increase system trust in automated driving. One reason for

TABLE III. RATING OF THE INFORMATION PERCEPTION OF THE HUD AND ARHUD [N=30]

Statements on Information Perception	HMI Concept	Mdn
1. How helpful was the visualization in the (AR)HUD for communicating the status of the automation?	HUD	13
	ARHUD	13
2. I think it is difficult to understand which situation ^a is associated with which visualization in the (AR)HUD.	HUD	3
	ARHUD	3
3. I think the quality of the displayed information in the (AR)HUD was helpful in understanding the situation ^a .	HUD	13
	ARHUD	12.5
4. I think the quantity of displayed information in the (AR)HUD was helpful in understanding the situation ^a .	HUD	12
	ARHUD	11.5
5. I have to remember a lot of details to understand the visualization in the (AR)HUD.	HUD	2
	ARHUD	3
6. I can easily find relevant information in the (AR)HUD.	HUD	12.5
	ARHUD	13
7. I can better monitor the situations ^a using the (AR)HUD.	HUD	13
	ARHUD	13
8. I think it makes sense to show the planned maneuver in the (AR)HUD.	HUD	15
	ARHUD	14
9. The information in the (AR)HUD was displayed at the correct time.	HUD	13
	ARHUD	12

^a. Situation means the actual and planned maneuver of the automated vehicle based on the present traffic situation.

the higher system trust with the ARHUD concept could be the more detailed and accurate presentation of the information in the ARHUD compared to the HUD. This would fit in with previous studies, which have shown that a more detailed and realistic representation of information can lead to higher trust [30, 31].

The usability of both HMI concepts was good, interpreted by [27]. The conclusion from this result is that the participants felt that both HMI concepts supported them effectively, efficiently and satisfactorily in monitoring the automated vehicle. Although no significant effect was found, ARHUD appears to support drivers better than HUD.

The participants indicated with a significant difference that they had to remember more details in the ARHUD than in the HUD. This result was also confirmed in the explorative survey, in which the presentation in ARHUD was partly perceived as overloaded in individual traffic situations. This appears to indicate a partly too high information density in ARHUD, which increases the danger of concealing real objects as well as the cognitive stress in interpreting the visualization. One possible reason for this could be the simultaneous visualization of several elements, especially in complex traffic scenarios. In future studies, care should therefore be taken to ensure that less information is displayed simultaneously in the ARHUD.

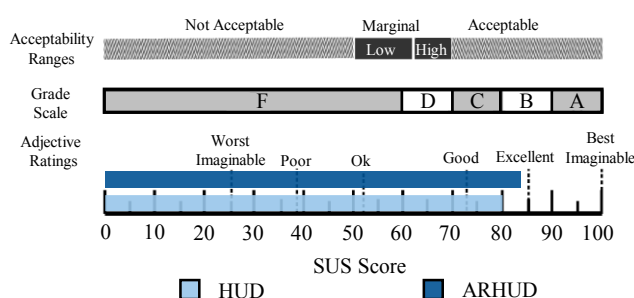


Fig. 5. Average SUS score of the concepts HUD (80.17) and ARHUD (83.58) [n = 30]

VI. CONCLUSION

Considering the results, the ARHUD concept has advantages in comparison to the HUD concept for system and scenario monitoring in urban partially automated driving in respect of subjective workload, trust, and gaze behavior, where the gaze can be directed to the relevant or critical areas in the field of view. Due to these advantages, we propose a combination of ARHUD, HUD and IC for further investigations in urban partially automated driving. As applied in this study, a HUD could be useful as a preview in situations where a correct local display is only possible at a late stage in the ARHUD. In conclusion, the controllability, if there are function limits in respect of the partial automation, should be evaluated supported by the developed concept. In addition, it should be ensured that the high level of trust when using an ARHUD does not lead to overreliance on it during use, which could negatively affect performance upon take-over of control by the driver.

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